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NeuroEdge: RISC-V SoC with Systolic-Array AI Acceleration

Comprehensive Architecture, Complete Signal & Register Catalog, Memory Maps, and Fault Tolerances

1. Introduction

1.1 System Overview

NeuroEdge is a domain-specific, 32-bit RISC-V System-on-Chip (SoC) engineered for low-power, deterministic neural-network inference at the edge. Edge and IoT deployments—such as wake-word detection, vibration anomaly identification, biometric classification, and sensor-fusion tasks—require localized execution of deep learning models. Performing inference directly on-chip eliminates latency overheads, preserves raw telemetry privacy, and circumvents continuous RF energy expenditure.

The dominant computational cost in modern edge neural networks is matrix-vector (M × V) and matrix-matrix (M × M) Multiply-Accumulate (MAC) calculations within dense fully-connected (FC) and 2D convolutional (Conv2D) layers. Traditional general-purpose embedded microcontroller cores execute these nested arithmetic loops sequentially in software, which is energy-inefficient and consumes cycles needed for real-time control, scheduling, and protocol management. NeuroEdge solves this computational bottleneck by integrating a 2D hardware systolic-array inference accelerator driven by an autonomous SPI-fed Direct Memory Access (DMA) streaming pipeline.

1.2 Purpose

The primary purpose of the NeuroEdge SoC is to provide a complete, verified, memory-mapped inference engine that executes quantized neural network layers while maintaining low power consumption suitable for battery-operated smart nodes, industrial condition monitors, and wearables. The architecture isolates compute-intensive matrix arithmetic into dedicated hardware, reserving the central processor strictly for system boot, peripheral orchestration, Flash weight retrieval, DMA descriptor setup, and lightweight non-linear activation post-processing.

1.3 Design Goals

- **Offloaded Execution Plane:** Zero CPU intervention during systolic matrix-multiplication passes; the processor configures DMA transfers and enters a low-power sleep state until alerted by completion interrupts.
- **High MAC Throughput per Watt:** A two-dimensional 4×4 grid of Processing Elements (PEs) executing 16 parallel signed 8-bit MAC operations per clock cycle with localized partial-sum forwarding.
- **Autonomous Parameter Streaming:** A master SPI controller that reads quantized INT8 weights from external SPI NOR Flash directly into Data SRAM without processor bit-banging.
- **Deterministic Scheduling & Safety:** A 32-bit programmable system timer for periodic inference scheduling combined with a hardware Watchdog Timer (WDT) that enforces pipeline liveness and fault recovery.
- **Standardized Bus Interface:** Clean memory-mapped registers conforming to 32-bit pipelined Wishbone B4 / AXI4-Lite bus protocols.

1.4 Main Inputs and Outputs

- **Clock and Reset:** Global synchronized system clock (`clk` , 50–100 MHz) and master active-low synchronous reset (`rst_n`).
- **Flash Memory Interface:** `spi_sclk` (serial clock), `spi_mosi` (master-out), `spi_cs_n` (active-low chip select), and `spi_miso` (master-in weight stream).
- **Telemetry & Communications:** `uart_tx` (serial transmission of classification results and profiling data) and `uart_rx` (serial command ingestion from host).
- **General Purpose Digital I/O:** `gpio_in[7:0]` (external hardware push-buttons, sensor alert lines) and `gpio_out[7:0]` (visual indicators, pipeline-stage debug pins).
- **Watchdog Reset Line:** `wdt_reset` asserted into the system reset distribution tree.

1.5 Hardware and Software Components

- **Hardware Blocks (RTL Domain):** RV32I/IM CPU core, 32 KB Instruction SRAM, 64 KB Dual-Port Data SRAM, Shared 32-bit Interconnect Fabric, 4×4 Systolic-Array MAC Accelerator, Dual-Channel DMA Controller, Master SPI Flash Controller, 16550 UART Transceiver, 32-bit Precision Timer, Hardware Watchdog Timer, 8-bit GPIO Module, and Platform Interrupt Controller (PIC).
- **Software Blocks (Firmware Domain):** Hardware Abstraction Layer (HAL) drivers, SPI Flash parameter loader, DMA transaction manager, software activation function library (ReLU, Leaky ReLU, Softmax, ArgMax), and telemetry serialization routines.

2. Detailed System Architecture

2.1 Complete Hierarchical Architecture

The architecture partitions tasks between an administrative Control Plane and an autonomous Data Plane. The Control Plane runs supervisor software from Instruction SRAM, configuring peripheral registers via memory-mapped load/store instructions. The Data Plane streams quantized parameters and activations between memory and compute units via dedicated DMA channels without CPU cycle-stealing.

2.2 Subsystem Descriptions

- **RISC-V Core:** A 32-bit scalar, single-issue pipelined processor core executing the base integer instruction set (RV32I) with optional hardware multiplier/divider (RV32M). Operates in Machine-mode (M-mode) to handle trap entry, interrupt servicing, and system boot.
- **SPI Controller:** A master SPI interface running SPI Mode 0 (CPOL=0, CPHA=0) that communicates with external non-volatile Flash. Supports automatic read streaming into Data SRAM with configurable clock division.
- **DMA Controller:** A dual-channel memory-to-accelerator streaming engine. Channel 0 moves activation vectors from Data SRAM to the systolic array boundary registers; Channel 1 moves weight matrices into the stationary PE registers.
- **Systolic-Array MAC Accelerator:** A 2D grid of 16 Processing Elements arranged in a 4×4 topology. Implements a Weight-Stationary (WS) execution model where weights remain resident inside PEs while activations stream horizontally and partial sums accumulate vertically.
- **Bus Interconnect:** A 32-bit synchronous Wishbone B4 pipelined (or AXI4-Lite) interconnect fabric with round-robin arbitration, address decoding, and integrated bus-error timeout detection.
- **UART Controller:** Full-duplex asynchronous serial transceiver with independent 8-byte transmit and receive FIFOs and programmable 16-bit baud divisor.
- **Timer Subsystem:** Free-running 32-bit up-counter with a programmable match comparator for periodic inference triggering.
- **Watchdog Timer (WDT):** Safety-critical countdown timer requiring periodic software heartbeats (`wdt_kick`); forces an active-high reset pulse if the pipeline stalls.
- **Platform Interrupt Controller (PIC):** Aggregates 6 peripheral interrupt lines, providing prioritization, individual masking, and vectored routing to the CPU.

2.3 Processing Pipeline Description

1. **Stage 1 (Weight Staging):** The processor programs the SPI controller with Flash source address and transfer length. Weights stream from external Flash over `spi_miso` directly into Data SRAM (`0x2000_8000`).
2. **Stage 2 (Streaming into the Array):** The processor programs the DMA Controller. DMA Channel 1 pre-loads weights into the 16 PEs. DMA Channel 0 then streams quantized activations from Data SRAM (`0x2000_A000`) into the array boundary registers, qualified by `pe_valid` .
3. **Stage 3 (Systolic Computation):** The 4×4 PE mesh propagates activations eastwards and partial sums southwards. After 2N - 1 cycles, accumulated 32-bit results settle into boundary result registers, and the array asserts `result_valid` and raises `array_irq` .
4. **Stage 4 (Result Handling & I/O):** On `array_irq` , the core reads the result registers, computes non-linear activation functions (ReLU, Softmax) in software, and pushes classification labels off-chip via UART. The timer is armed for the next scheduled pass.

3. Detailed Block Diagram

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4. Complete Signal Path Analysis

SIGNAL NAME	SOURCE → DEST	WIDTH	DATA TYPE	TIMING / RATE	OPERATIONAL PURPOSE
`clk`	Global Pad / PLL	1 bit	Clock	50–100 MHz	Master synchronized system clock distribution tree.
`rst_n`	Pad / Reset Tree	1 bit	Control	Min 3 cycles	Active-low synchronous system reset to all sequential registers.
`bus_addr[31:0]`	CPU / DMA Master	32 bits	Address	Setup 2.0 ns	Memory-mapped address bus targeting memories and peripheral registers.
`bus_wdata[31:0]`	CPU / DMA Master	32 bits	Data Word	Coincident stb	Master parallel write data bus.
`bus_rdata[31:0]`	Slaves / Memory	32 bits	Data Word	Valid on ack	Slave parallel read response data bus returned to master.
`bus_we`	Bus Master	1 bit	Control	Address phase	Write enable flag: 1 = Write transaction, 0 = Read transaction.
`bus_stb`	Bus Master	1 bit	Control	Active transfer	Strobe line indicating valid address and transfer cycle.
`bus_cyc`	Bus Master	1 bit	Control	Full transfer	Cycle envelope active for entire bus transaction duration.
`bus_ack`	Addressed Slave	1 bit	Control	1-cycle pulse	Slave transfer acknowledge handshake terminating bus phase.
`spi_sclk`	SPI Controller	1 bit	Clock	Up to 25 MHz	Serial clock driven to external Flash memory.
`spi_mosi`	SPI Controller	1 bit	Serial Data	Falling SCLK	Master-out data line sending commands and addresses to Flash.
`spi_miso`	External Flash	1 bit	Serial Data	Rising SCLK	Master-in data line streaming weight bytes into the SoC.
`spi_cs_n`	SPI Controller	1 bit	Control	Active transfer	Active-low slave chip select asserting across Flash access frames.

SIGNAL NAME	SOURCE → DEST	WIDTH	DATA TYPE	TIMING / RATE	OPERATIONAL PURPOSE
`spi_start`	CPU (Register)	1 bit	Control	Pulse	Software-driven trigger initiating SPI Flash read transfer.
`spi_done`	SPI Controller	1 bit	Flag / IRQ	Pulse	Asserted when commanded Flash transfer length has completed.
`dma_src_addr[31:0]`	CPU (Register)	32 bits	Address	Pre-transfer	DMA base source address in Data SRAM.
`dma_dst_addr[31:0]`	CPU (Register)	32 bits	Address	Pre-transfer	DMA base destination address targeting array boundary port.
`dma_length[15:0]`	CPU (Register)	16 bits	Word Count	Pre-transfer	Total burst transfer length in 32-bit words.
`dma_start`	CPU (Register)	1 bit	Control	Pulse	Software strobe initiating DMA streaming transfer.
`dma_busy`	DMA Engine	1 bit	Status Flag	Continuous	Held high while a DMA block transfer is underway.
`dma_irq`	DMA Engine	1 bit	Interrupt	Level-held	Asserted upon DMA transfer word counter reaching zero.
`weight_data[7:0]`	DMA Controller	8 bits	INT8 Signed	Per PE cycle	Quantized model weight byte delivered to local PE registers.
`act_data[7:0]`	DMA Controller	8 bits	INT8 Signed	Per PE cycle	Quantized input activation byte fed into array western boundary.
`pe_valid`	DMA Controller	1 bit	Control	Synchronous	Valid handshake qualifying weight_data and act_data streams.
`array_start`	CPU (Register)	1 bit	Control	Pulse	Trigger signal initiating systolic compute pass.
`psum_n_s_e_w[31:0]`	PE Array Internal	32 bits	INT32 Signed	Array Internal	Inter-PE partial sum and activation forwarding interconnect links.
`result_data[31:0]`	Systolic Array	32 bits	INT32 Signed	Valid on done	Final accumulated dot-product output element.
`result_valid`	Systolic Array	1 bit	Status Flag	Level-held	Asserted when result_data output registers hold complete values.
`array_irq`	Systolic Array	1 bit	Interrupt	Level-held	Completion interrupt raised to PIC when matrix pass concludes.
`uart_tx`	UART Peripheral	1 bit	Serial Asynch	Baud rate clock	Serial transmit carrying classification labels and telemetry off-chip.
`uart_rx`	External Pad	1 bit	Serial Asynch	16x Baud sample	Serial receive line for host control commands.
`uart_busy`	UART Peripheral	1 bit	Status Flag	Active frame	Held high while UART shift register transmits a frame.
`uart_irq`	UART Peripheral	1 bit	Interrupt	Level-held	Asserted on TX FIFO empty or RX byte arrival.
`timer_cnt[31:0]`	Timer Subsystem	32 bits	Counter	Continuous	Free-running 32-bit system counter value.
`timer_cmp[31:0]`	CPU (Register)	32 bits	Threshold	Pre-set	Compare register triggering periodic timer interrupts.

SIGNAL NAME	SOURCE → DEST	WIDTH	DATA TYPE	TIMING / RATE	OPERATIONAL PURPOSE
`timer_irq`	Timer Subsystem	1 bit	Interrupt	Level-held	Asserted when timer_cnt matches timer_cmp.
`gpio_in[7:0]`	External Pads	8 bits	Digital Logic	Asynchronous	External hardware inputs synchronized via 2-stage flip-flops.
`gpio_out[7:0]`	GPIO Module	8 bits	Digital Logic	Synchronous	Status indicators and execution profiling pulses.
`gpio_dir[7:0]`	CPU (Register)	8 bits	Control Mask	Pre-set	Per-pin direction mask: 0 = Input, 1 = Output.
`wdt_kick`	CPU / Heartbeat	1 bit	Pulse	Periodic	Liveness refresh strobe that reloads the Watchdog counter.
`wdt_timeout[31:0]`	CPU (Register)	32 bits	Cycle Count	Pre-set	Programmed timeout interval in clock cycles.
`wdt_reset`	Watchdog Timer	1 bit	Reset Pulse	Min 16 cycles	Global system reset pulse asserted upon missed watchdog kick.
`irq_vec[5:0]`	PIC Subsystem	6 bits	Interrupt Vector	Level-held	Prioritized interrupt lines routed to the RISC-V trap handler.

5. Data Path Analysis

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The data path executes quantized INT8 matrix-vector multiplications using a Weight-Stationary structure across 16 Processing Elements:

- **Weight Pre-loading:** Prior to computation, weights are read from the SRAM staging region and written into local registers inside each PE cell.
- **Skewed Activation Delivery:** Activations are injected into array rows with a 1-cycle stagger: Cycle 0 receives Row 0 ($A_{0,0}$); Cycle 1 receives Row 0 ($A_{0,1}$) and Row 1 ($A_{1,0}$); Cycle 2 receives Row 0 ($A_{0,2}$), Row 1 ($A_{1,1}$), and Row 2 ($A_{2,0}$). This aligns horizontal movement with vertical accumulation.
- **Arithmetic Execution:** Each PE computes $P_{out} = P_{in} + (A_{in} \times W_{local})$. Multiplications operate on signed 8-bit integers; partial sums accumulate in 32-bit signed registers.
- **Accumulator Readout:** After 2N - 1 cycles (7 clock cycles for $N=4$), final accumulated values settle into result registers, triggering `array_irq`.

6. Control Path Analysis

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[ System Power-On / Reset (rst_n = 0) ] | v [ RISC-V Boot: Set SP, Register Trap Vector mtvec, Clear BSS ] | v [
```

7. Interrupt Architecture

The Platform Interrupt Controller (PIC) arbitrates between 6 peripheral interrupt lines and maps them directly to the core's `irq_vec[5:0]` input port:

INTERRUPT SOURCE	PRIORITY	TRIGGER CONDITION	HARDWARE ISR ACTION	RECOVERY & CLEARING
Watchdog Alert	0 (Highest)	Counter reaches zero without a heartbeat.	Dumps processor context; logs fault telemetry.	Write `0xA5A5_5A5A` to `WDT_KICK`.
Timer Tick	1	`TIMER_CNT == TIMER_CMP`	Schedules periodic inference run; updates system ticks.	Write 1 to clear `TIMER_STATUS.MATCH`.
Systolic Array Done	2	Matrix dot-product accumulation complete.	Reads result registers; runs Softmax/ArgMax in software.	Write 1 to clear `ARRAY_STATUS.DONE`.
DMA Complete	3	Transfer length decrements to zero.	Releases memory locks; advances ring descriptors.	Write 1 to clear `DMA_STATUS.DONE`.
UART Complete	4	TX FIFO empty / RX byte ready.	Pushes remaining telemetry bytes or reads host command.	Write byte to `UART_TX_DATA`.
GPIO Trigger	5 (Lowest)	Configured edge/level detected on input pins.	Processes external push-button or wake signal.	Write 1 to clear `GPIO_STATUS.FLAG`.

8. Complete Register Architecture

All registers are 32-bit wide, aligned to 4-byte boundaries, with support for Read (R), Write (W), Read/Write (R/W), and Write-1-to-Clear (W1C) access modes.

8.1 SPI Controller Registers (Base: 0x4000_0000)

OFFSET / REG	BITFIELDS	ACCESS	RESET	DESCRIPTION & OPERATIONAL LOGIC
SPI_CTRL `0x00`	[0] EN [1] START [2] AUTO_READ [31:3] RESERVED	R/W R/W R/W R	`0x0000_0000`	Module enable; one-shot transfer start trigger; auto-stream read payload directly to Data SRAM.
SPI_STATUS `0x04`	[0] BUSY [1] DONE [2] RX_EMPTY [31:3] RESERVED	R W1C R R	`0x0000_0004`	Transfer active; byte/block transfer completed; receive FIFO empty flag.
SPI_CLKDIV `0x08`	[7:0] DIVISOR [31:8] RESERVED	R/W R	`0x0000_0004`	Clock prescaler: $f_{SCLK} = f_{CLK} / (2 \times DIVISOR)$. Default yields 12.5 MHz at 50 MHz system clock.
SPI_FLASH_ADDR `0x0C`	[23:0] FLASH_ADDR [31:24] CMD_BYTE	R/W R/W	`0x0300_0000`	24-bit base Flash memory byte address; SPI command instruction byte (default 0x03 Standard Read).
SPI_LENGTH `0x10`	[15:0] BYTE_LEN [31:16] RESERVED	R/W R	`0x0000_0000`	Programmed read burst length in bytes (e.g., 16 to 1024 bytes per layer matrix).

8.2 DMA Controller Registers (Base: 0x4000_1000)

OFFSET / REG	BITFIELDS	ACCESS	RESET	DESCRIPTION & OPERATIONAL LOGIC
DMA_CTRL `0x00`	[0] DMA_EN [1] CH0_EN [2] CH1_EN [31:3] RESERVED	R/W R/W R/W R	`0x0000_0000`	Global DMA enable; Channel 0 enable (Activation stream); Channel 1 enable (Weight stream).
DMA_STATUS `0x04`	[0] BUSY [1] CH0_DONE [2] CH1_DONE [8] ERR	R W1C W1C W1C	`0x0000_0000`	Channel transfer in progress; Channel 0 block complete; Channel 1 block complete; bus access/alignment error.

OFFSET / REG	BITFIELDS	ACCESS	RESET	DESCRIPTION & OPERATIONAL LOGIC
DMA_CH0_SRC `0x10`	[31:0] SRC_ADDR	R/W	`0x0000_0000`	Channel 0 source base address in Data SRAM (Activation staging buffer).
DMA_CH0_DST `0x14`	[31:0] DST_ADDR	R/W	`0x4000_2014`	Channel 0 destination address targeting array activation boundary port (`ARRAY_ACT_PORT`).
DMA_CH0_LEN `0x18`	[15:0] XFER_LEN [31:16] RESERVED	R/W R	`0x0000_0000`	Channel 0 transfer size in 32-bit words.
DMA_CH1_SRC `0x20`	[31:0] SRC_ADDR	R/W	`0x0000_0000`	Channel 1 source base address in Data SRAM (Weight staging buffer).
DMA_CH1_DST `0x24`	[31:0] DST_ADDR	R/W	`0x4000_2010`	Channel 1 destination address targeting array weight loading port (`ARRAY_WEIGHT_PORT`).
DMA_CH1_LEN `0x28`	[15:0] XFER_LEN [31:16] RESERVED	R/W R	`0x0000_0000`	Channel 1 transfer size in 32-bit words.

8.3 Systolic-Array MAC Accelerator Registers (Base: 0x4000_2000)

OFFSET / REG	BITFIELDS	ACCESS	RESET	DESCRIPTION & OPERATIONAL LOGIC
ARRAY_CTRL `0x00`	[0] EN [1] START [2] CLEAR_ACC [31:3] RESERVED	R/W R/W W1C R	`0x0000_0000`	Accelerator master enable; software computation start strobe; clear internal 32-bit accumulators to zero.
ARRAY_STATUS `0x04`	[0] BUSY [1] DONE [2] OVERFLOW [31:3] RESERVED	R W1C W1C R	`0x0000_0000`	Systolic mesh computing; computation pass complete; 32-bit partial-sum signed arithmetic overflow flag.
ARRAY_CFG `0x08`	[1:0] MODE [7:4] DIM [31:8] RESERVED	R/W R/W R	`0x0000_0040`	Execution mode: 00 = Weight Stationary; active array dimension (default 4 for 4×4 mesh).
ARRAY_WEIGHT_PORT `0x10`	[31:0] WGT_WORD	W	`0x0000_0000`	Memory-mapped parallel write port for loading packed INT8 weights into PE stationary registers.
ARRAY_ACT_PORT `0x14`	[31:0] ACT_WORD	W	`0x0000_0000`	Memory-mapped parallel write port for streaming packed INT8 activations into the array western boundary.
ARRAY_RES_0 `0x20`	[31:0] RES_COL_0	R	`0x0000_0000`	Accumulated 32-bit signed dot-product output for Column 0.
ARRAY_RES_1 `0x24`	[31:0] RES_COL_1	R	`0x0000_0000`	Accumulated 32-bit signed dot-product output for Column 1.
ARRAY_RES_2 `0x28`	[31:0] RES_COL_2	R	`0x0000_0000`	Accumulated 32-bit signed dot-product output for Column 2.
ARRAY_RES_3 `0x2C`	[31:0] RES_COL_3	R	`0x0000_0000`	Accumulated 32-bit signed dot-product output for Column 3.

8.4 UART Registers (Base: 0x4000_3000)

OFFSET / REG	BITFIELDS	ACCESS	RESET	DESCRIPTION & OPERATIONAL LOGIC
UART_TX_DATA `0x00`	[7:0] TX_BYTE [31:8] RESERVED	W R	`0x0000_0000`	Write byte to Transmit FIFO; transmission initiates automatically when transmitter is idle.
UART_RX_DATA `0x04`	[7:0] RX_BYTE [31:8] RESERVED	R R	`0x0000_0000`	Read byte from Receive FIFO; pops the oldest byte from buffer.
UART_STATUS `0x08`	[0] TX_BUSY [1] TX_FULL [2] RX_VALID [3] RX_OVERRUN	R R R W1C	`0x0000_0000`	Transmit active; TX FIFO full; byte available in RX FIFO; receiver hardware overrun error.
UART_BAUD_DIV `0x0C`	[15:0] DIVISOR [31:16] RESERVED	R/W R	`0x0000_01B2`	Baud rate prescaler: DIV = f_{CLK} / (16 × Baud). Default 0x01B2 (434) yields 115200 bps at 50 MHz.

8.5 Timer Registers (Base: 0x4000_4000)

OFFSET / REG	BITFIELDS	ACCESS	RESET	DESCRIPTION & OPERATIONAL LOGIC
TIMER_CTRL `0x00`	[0] EN [1] AUTO_RELOAD [31:2] RESERVED	R/W R/W R	`0x0000_0000`	Counter enable; auto-reload counter from zero upon compare match.
TIMER_CNT `0x04`	[31:0] COUNTER	R/W	`0x0000_0000`	Current 32-bit free-running counter value; increments on every system clock cycle.
TIMER_CMP `0x08`	[31:0] COMPARE	R/W	`0xFFFF_FFFF`	Compare threshold value; asserts `timer_irq` when `TIMER_CNT == TIMER_CMP`.
TIMER_STATUS `0x0C`	[0] MATCH_FLAG [31:1] RESERVED	W1C R	`0x0000_0000`	Asserted on compare match; cleared by writing 1.

8.6 Watchdog Timer Registers (Base: 0x4000_5000)

OFFSET / REG	BITFIELDS	ACCESS	RESET	DESCRIPTION & OPERATIONAL LOGIC
WDT_CTRL `0x00`	[0] EN [1] RESET_EN [31:2] RESERVED	R/W R/W R	`0x0000_0001`	Watchdog master enable; assert hardware reset signal upon counter expiration.
WDT_KICK `0x04`	[31:0] KEY	W	`0x0000_0000`	Write magic value `0xA5A5_5A5A` to reload watchdog countdown counter.
WDT_TIMEOUT `0x08`	[31:0] TIMEOUT	R/W	`0x004C_4B40`	Timeout window in system clock cycles (default 5,000,000 cycles = 100 ms at 50 MHz).
WDT_STATUS `0x0C`	[0] BARK_FLAG [31:1] RESERVED	W1C R	`0x0000_0000`	Asserted when timer expires without a kick; cleared by writing 1.

8.7 GPIO Registers (Base: 0x4000_6000)

OFFSET / REG	BITFIELDS	ACCESS	RESET	DESCRIPTION & OPERATIONAL LOGIC
GPIO_IN `0x00`	[7:0] DATA_IN [31:8] RESERVED	R R	`0x0000_0000`	Synchronized logic levels sampled on external GPIO pins.
GPIO_OUT `0x04`	[7:0] DATA_OUT [31:8] RESERVED	R/W R	`0x0000_0000`	Output register driving external pins configured as outputs.
GPIO_DIR `0x08`	[7:0] PIN_DIR [31:8] RESERVED	R/W R	`0x0000_0000`	Direction mask: 0 = Input, 1 = Output (default all inputs).

8.8 Interrupt Controller (PIC) Registers (Base: 0x4000_7000)

OFFSET / REG	BITFIELDS	ACCESS	RESET	DESCRIPTION & OPERATIONAL LOGIC
PIC_ENABLE `0x00`	[5:0] INT_MASK [31:6] RESERVED	R/W R	`0x0000_0000`	Interrupt enable mask (Bit 0 = WDT, 1 = Timer, 2 = Array, 3 = DMA, 4 = UART, 5 = GPIO).
PIC_PENDING `0x04`	[5:0] INT_PEND [31:6] RESERVED	R R	`0x0000_0000`	Raw pending interrupt flags from peripheral lines before masking.
PIC_PRIORITY `0x08`	[23:0] PRIO_LEVELS [31:24] RESERVED	R/W R	`0x0054_3210`	Nibble-encoded priority assignments (0 = highest, 5 = lowest).
PIC_ACK `0x0C`	[5:0] INT_ACK [31:6] RESERVED	W1C R	`0x0000_0000`	Write 1 to clear pending interrupt latch in PIC arbiter.

9. Memory Architecture

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0xFFFF_FFFF +-------------------------------------------------------+ | Reserved High Memory Space | 0x4000_
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ADDRESS RANGE	SIZE	MASTER ACCESS	REGION TYPE	FUNCTIONAL DESCRIPTION
`0x0000_0000 - 0x0000_7FFF`	32 KB	CPU Fetch	Instruction SRAM	Bootloader, firmware code segment (`.text`), and vector table.
`0x2000_0000 - 0x2000_7FFF`	32 KB	CPU Load/Store	Data SRAM (Stack)	Application stack (grows downwards from 0x2000_7FE0) and heap.
`0x2000_8000 - 0x2000_9FFF`	8 KB	CPU, DMA	Data SRAM (Weights)	Quantized INT8 model parameter staging area loaded from Flash.
`0x2000_A000 - 0x2000_AFFF`	4 KB	CPU, DMA	Data SRAM (Acts)	Input activation vectors queued for systolic processing passes.
`0x2000_B000 - 0x2000_FFFF`	20 KB	CPU	Data SRAM (Outs)	Accumulated output vectors, intermediate feature maps, and UART buffers.
`0x4000_0000 - 0x4000_001F`	32 B	CPU	SPI Registers	SPI Flash control, status, divisor, address, and length registers.
`0x4000_1000 - 0x4000_102F`	48 B	CPU	DMA Registers	Channel source/destination addresses, transfer counts, and controls.
`0x4000_2000 - 0x4000_203F`	64 B	CPU, DMA	Systolic Registers	Array control, PE streaming ports, and 32-bit output result registers.
`0x4000_3000 - 0x4000_301F`	32 B	CPU	UART Registers	UART baud divisor, TX/RX FIFOs, and transmission status flags.
`0x4000_4000 - 0x4000_401F`	32 B	CPU	Timer Registers	32-bit counter, compare threshold, and periodic trigger status.
`0x4000_5000 - 0x4000_501F`	32 B	CPU	Watchdog Registers	Heartbeat reload port, timeout setting, and status indicators.
`0x4000_6000 - 0x4000_600F`	16 B	CPU	GPIO Registers	8-bit input sampling, output drive registers, and direction masks.
`0x4000_7000 - 0x4000_701F`	32 B	CPU	PIC Registers	Interrupt enable mask, pending register, priority map, and ACK port.

10. Error Handling Architecture

- **Hardware Watchdog Reset:** If a DMA streaming transaction or systolic array pass stalls due to dropped handshakes, the WDT counter reaches zero and asserts an active-high `wdt_reset` pulse, executing a clean hardware reboot.
- **Bus Misalignment & Contention:** The interconnect arbiter enforces a 64-cycle timeout; transactions failing to receive `bus_ack` trigger a precise Bus Error trap (`mcause = 0x5`). DMA transfers targeting non-4-byte aligned boundaries immediately flag `DMA_STATUS.ERR` and halt.
- **Accumulator Overflow Protection:** Each PE incorporates signed saturation logic that clamps 32-bit partial sums at `0x7FFF_FFFF` (positive) or `0x8000_0000` (negative), setting `ARRAY_STATUS.OVERFLOW` while preventing arithmetic wrap-around anomalies.
- **UART Overrun:** The UART receiver asserts `UART_STATUS.RX_OVERRUN` if bytes arrive when the internal 8-byte FIFO is full, preserving prior data integrity.
- **Fail-Safe Behavior:** During critical faults, external outputs are clamped to known inactive states, `uart_tx` is held high (idle), and `gpio_out[7:0]` drives an error diagnostic pattern (`0xEE`).

11. Operating Modes

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+------------------+ | STARTUP MODE | <----+ (Cold / Warm Reset) +--------+---------+ | | | v | +---------------
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- **Startup Mode:** Bootloader initializes system stacks, sets trap handlers, verifies SRAM BIST, and arms the watchdog heartbeat.
- **Configuration Mode:** The core commands the SPI master to load model weights from Flash into SRAM and configures DMA descriptors.
- **Normal Processing Mode:** DMA streams activations into the systolic grid, matrix-multiplication runs in hardware, and the CPU evaluates layer post-processing.
- **Idle Mode:** Clocks to the systolic array and DMA are dynamic-gated; the CPU enters the low-power `wfi` sleep state until awakened by `timer_irq` .
- **Error Mode:** Fault conditions halt the pipeline, clamp outputs, and broadcast diagnostic telemetry over UART.
- **Recovery Mode:** Software flushes FIFOs, asserts soft-reset bits, and re-reads model weights from Flash.
- **Shutdown Mode:** All peripheral clocks are gated, and SRAM is held in low-leakage state-retention mode.

12. Interface Specifications

- **Shared Interconnect:** Wishbone B4 pipelined / AXI4-Lite compatible, 32-bit data/address, single-cycle acknowledgment, setup 2.0 ns, hold 0.5 ns.
- **SPI Master:** SPI Mode 0 (CPOL=0, CPHA=0), serial clock frequency up to 25 MHz, 4 pins (`spi_sclk` , `spi_mosi` , `spi_miso` , `spi_cs_n`).
- **Systolic Streaming Interface:** Point-to-point streaming handshake (`weight_data[7:0]` , `act_data[7:0]` , qualified by synchronous `pe_valid`).
- **UART Transceiver:** Asynchronous serial framing (1 Start, 8 Data, No Parity, 1 Stop bit) at 115200–921600 bps.
- **GPIO Lines:** 8-bit bidirectional digital I/O with 2-stage input synchronizers.

13. Target Applications

- **Audio Wake-Word Detection:** Evaluates quantized 1D depthwise separable convolutional layers on 16 kHz audio features (MFCCs) with sub-5 ms inference latency.
- **Industrial Vibration Anomaly Detection:** Evaluates multi-layer perceptron (MLP) layers on 3-axis accelerometer data to classify bearing fault harmonics.
- **Wearable Gesture Recognition:** Classifies real-time 6-axis IMU motion trajectories directly on-device.
- **Low-Power Edge Vision:** Executes visual wake-word convolutional kernels (3×3 Conv2D) to detect human presence.

14. Design Assumptions

1. A verified 32-bit RV32I/IM core with memory-mapped load/store buses and external interrupt lines is provided as an integrated module.
2. Weights and activations are uniformly quantized to signed 8-bit integers (INT8), with 32-bit signed accumulators.
3. The systolic array is configured as a 4×4 grid operating in a Weight-Stationary structure; larger matrices are tiled in software.
4. The SoC operates within a single synchronous clock domain (`clk`).
5. External Flash is pre-programmed with quantized neural network parameters starting at address 0x000000.

15. Design Limitations

1. Matrix dimensions must be zero-padded to multiples of 4 by software before execution.
2. On-chip storage is capped at 32 KB Instruction SRAM and 64 KB Data SRAM; larger neural networks must be partitioned across Flash.
3. Non-linear activation functions (Softmax, ArgMax) and bias additions are evaluated in software by the RISC-V core.
4. Concurrent access by CPU and DMA on the shared interconnect can introduce minor wait-states during activation streaming passes.

16. Verification Strategy

Verification follows a multi-tier SystemVerilog / UVM methodology:

- **Module Verification:** The systolic array is stimulated with randomized INT8 matrices and checked against a bit-true NumPy reference model. The SPI controller is validated against a behavioral Flash model.
- **Integration Verification:** Confirms zero dropped beats or pipeline bubbles across the DMA-to-array streaming interface under bus backpressure.
- **System-Level Testbench:** A complete neural-network classifier (MNIST or wake-word model) is simulated from reset: weights are loaded from simulated Flash, activations are streamed via DMA, partial sums propagate, and final UART output packets are evaluated for 100% numerical match.
- **Coverage Target:** > 95% statement and branch coverage, with 100% transition coverage on all FSMs.

17. Final Consolidated Architecture Diagram

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